

Module 6: Metabolism - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. What is the main role of ATP in a cell?
2. A. To store long-term genetic information
3. B. To carry and deliver usable energy for cellular work
4. C. To act as a structural membrane
5. D. To speed up reactions as an enzyme
6. Answer: B

7. Why: ATP is the cell's energy currency: it couples energy-releasing reactions to energy-requiring ones.

8. How does an enzyme speed up a chemical reaction?
9. A. By lowering the activation energy needed to start it
10. B. By adding heat to the whole cell
11. C. By becoming a permanent part of the product
12. D. By raising the activation energy
13. Answer: A

14. Why: Enzymes lower activation energy. They are catalysts and are not used up in the reaction.

15. Raising temperature far above normal can stop an enzyme from working because:
16. A. all substrates suddenly disappear
17. B. ATP levels increase
18. C. the reaction runs backward forever
19. D. the enzyme denatures and loses its active-site shape
20. Answer: D

21. Why: Excess heat denatures the enzyme. Without its shape, the substrate no longer fits the active site.

22. How are photosynthesis and cellular respiration related?

23. A. They are exactly the same reaction.

24. B. Neither one uses energy.

25. C. Photosynthesis stores energy in sugars, and respiration releases it.

26. D. Both occur only in the dark.

27. Answer: C

28. Why: Photosynthesis builds sugars using light energy, and respiration harvests that stored energy. The two processes are complementary.